1. (40 points) Follow the guidelines for sketching a curve to draw the graph of the function

$$y = 2x + 3x^{2/3}.$$

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2. (40 points) A poster of area 6000 cm^2 has blank margins of width 10 cm on the top and bottom and 6 cm on the sides. Find the dimensions of the poster that maximize the printed area.

3. (60 points) Find the following integrals:

a) $\int_{-2}^{0} |1 + 2x|\, dx$ **b)** $\int \frac{\sin(\ln x)}{x}\, dx$ **c)** $\int_{1}^{2} 2xe^{x^2}\, dx$

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4. (40 points)

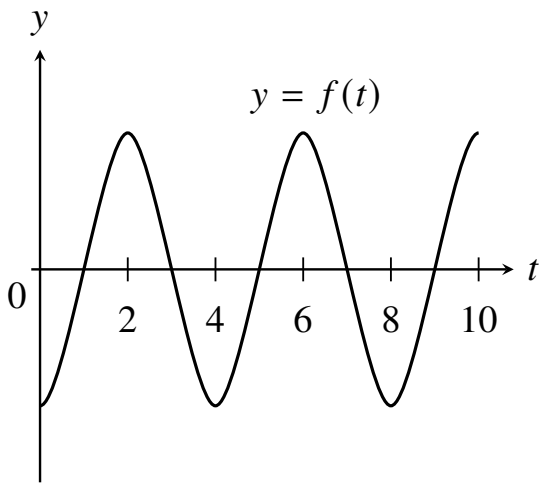
Consider the function f and its graph given in the picture.

a) Estimate the zeros of the area function

$$A(x) = \int_0^x f(t)\,dt, \qquad 0 \le x \le 10.$$

b) Estimate the points (if any) at which $A(x)$ has a local maximum or minimum.

c) Sketch a graph of $A(x)$, $0 \le x \le 10$, without a scale on the y-axis.



5. (50 points)

a) Find the volume obtained by rotating the region enclosed by the graph of $y = x^2$, the y-axis and the lines $y = 12 - x$ around the line $y = -2$ by using the method of disk and washer.

b) Find the volume of the solid obtained by rotating the region underneath the graph of $y = x^{-4}$ over the interval $[-3, -1]$ about the line $x = 4$ by using the method of cylindrical shells.

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6. (40 points) Find the area under the curve $y = x^2 + 2$ over the interval $[-3, -1]$ by computing the limit of the Riemann sum using the right endpoint of each subinterval.

7. (40 points) Find the following limits:

a) $\lim_{x \to 1^+} \left(\frac{3}{\ln x} - \frac{2}{x-1} \right)$

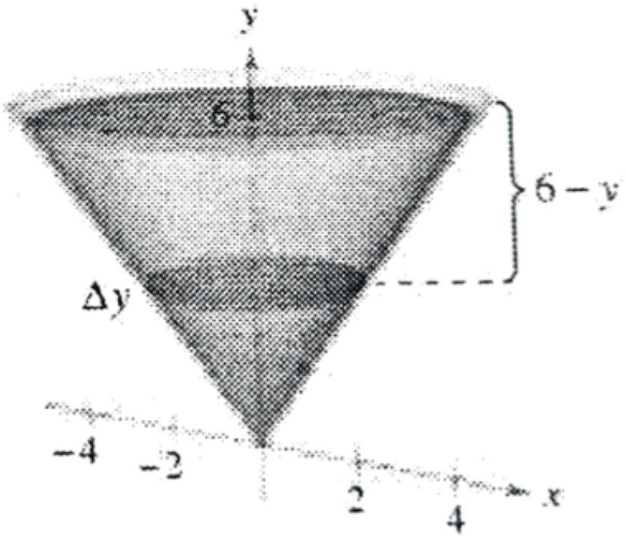
b) $\lim_{x \to 0^+} \left(\cos\left(\frac{\pi}{2} - x \right) \right)^x$

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8. (50 points) Water is pumped from the bottle of an empty tank in the figure below. How much work is done to fill the tank in each of the following cases:

a) To the water level of 2 feets.

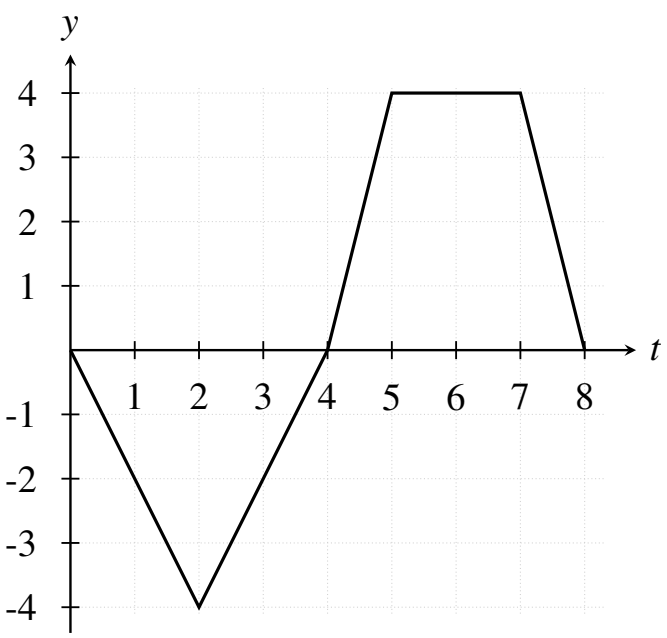
b) From the water level of 4 feets to the level of 6 feets.



9. (40 points) Given below is the graph of a function f(x) defined on the interval $[0, 6]$. Find the following:

a) $\int_2^5 f(x)\,dx$

b) $\int_6^0 f(x)\,dx$



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